

Data Cleaning & ETL

INSTANT

Pivot & Unpivot

Data Analysis Diploma • Session 06

PIVOT TABLES & POWER QUERY

Data Cleaning · Column Transformations · Pivot/Unpivot · Charts & Visual
Slicers

6 Goals**Core Outcomes**

ETL • Pivots • Charts

44 Slides**Curriculum Depth**

Power Query & Analytics

Visual Labs**Hands-on Practice**

Pivots & Slicers Worksheets

Pivot Tables & Group By

Start Session 06 →

Charts & Visual Slicers

— SESSION COMPETENCIES

What You Will Master Today

By the end of this session, you will master these advanced ETL, summarization, and chart analytics capabilities:



01

Advanced Data Cleaning

Eliminate duplicates, replace inconsistent text, and handle missing values in Power Query.



02

Reshape & Transform Columns

Split columns, merge fields, and extract structured substrings with automated steps.



03

Append & Merge Queries

Stack multi-month sales data and join relational lookup tables via primary/foreign keys.



04

Pivot & Unpivot Operations

Convert wide datasets to unpivoted long formats for flexible analysis and vice versa.



05

Dynamic Pivot Tables & Group By

Aggregate large datasets instantaneously, calculate custom metrics, and group by dates.

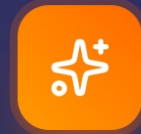


06

Pivot Charts & Slicer Dashboards

Build interactive dashboards using Column, Line, Scatter, Slicers, and Timelines.

Data Cleaning, Column Reshaping, Schema Alignment, Append & Merge Queries



PART 1

Power Query Advanced ETL

Data Cleaning, Column Reshaping, Schema Alignment, Append & Merge
Queries

— PIPELINE ARCHITECTURE

Power Query Workflow Review

Understanding where transformations and query combinations happen in the data engine:



01

1. Connect

Import raw data from Excel workbooks, CSV files, SQL databases, or web sources.



02

2. Transform

Clean data, split/merge columns, format data types, and apply filtering logic.



03

3. Combine

Stack matching tables (Append) or join related datasets via keys (Merge).



04

4. Load

Send refined analytical tables directly into Excel sheets or Data Model.

DATA SANITATION

Core Data Cleaning Operations

Tidying raw data before analysis to guarantee high-integrity reporting:



01

Remove Duplicates

Deletes identical rows based on selected key columns. Menu Path: Home > Remove Rows > Remove Duplicates.



02

Replace Values

Standardizes inconsistent text entries across a column. Menu Path: Transform > Replace Values.



03

Handle Missing Values

Fill down, fill up, or filter out incomplete null rows. Menu Path: Transform > Fill | Home > Remove Rows.

— STRUCTURAL REFORMATTING

Transforming Column Structure

Reshaping text fields to isolate key variables for analytical grouping:



01

Split Column

Break one column into several by delimiter (space, comma, hyphen), position, or character count.



02

Merge Columns

Combine two or more text columns into a single field using custom separators (e.g. Space, Hyphen).



03

Extract Text

Pull specific characters (e.g. first 3 chars of "SKU-2049" -> "SKU") or text before/after delimiter.

— TYPE INTEGRITY

Data Types & Why They Matter

Enforcing the proper data type on every column ensures calculations work as expected:



01

Text (ABC)

Used for names, codes, SKU numbers, and category labels where mathematical ops are not needed.



02

Whole Number / Decimal (123 / 1.2)

Numeric quantities, revenue, profit, and ratios. Enables sum, average, and formula math.



03

Date / Date-Time

Timestamps and transaction dates. Crucial for timeline filters, quarter grouping, and trend charts.



04

True / False (Boolean)

Binary flags (Active/Inactive, In-Stock/Out). Enables logical filtering and conditional rules.

— DATA TYPE CORRECTION

The Danger of Text-Stored Dates

How incorrect data types cause reporting features and date slicers to silently fail:



01

The Problem: "Order Date" as Text

When dates are imported as text, Excel treats them as literal strings. Date filters, year grouping, timeline slicers, and month-over-month formulas fail completely.



02

The Fix: Transform > Data Type > Date

In Power Query, select the column header -> click Transform tab -> Data Type -> change to Date. Results: timeline slicers, date sorting, and year/quarter aggregations work instantly!

— VERTICAL CONCATENATION

Append Queries: Stacking Datasets

Combining multiple data tables with identical column structures into a single continuous query:



01

How Append Works

Appends rows from Table B beneath Table A. Example: Jan_Sales (124 rows) + Feb_Sales (118 rows) = Combined_Sales (242 rows). Ideal for monthly regional files.



02

Key Requirement: Column Name Match

Columns must have exact matching headers (case-sensitive). If column names differ (e.g. "Amount" vs "SalesAmount"), Power Query creates separate mismatched columns.

— DATA QUALITY

Rules for Successful Append Queries

Guidelines for stacking multi-file regional data without introducing null artifacts:



01

Identical Column Names

Ensure exact matching names across files before appending. Rename mismatched headers in staging queries.



02

Consistent Data Types

Set identical data types across all source queries to avoid type casting conflicts during load.



03

Automatic Folder Import

Use Power Query "From Folder" connector to automatically append new monthly CSV/Excel files as they drop.

— HORIZONTAL COMBINATION

Merge Queries: Joining Tables on Shared Keys

Combining columns from two related tables based on a matching key ID field:



01

How Merge Works

Pairs rows from Orders table with Customer_Lookup table using a shared key like CustomerID. Orders gains Customer Name, Region, and Segment columns.



02

Primary & Foreign Keys

The key field must contain matching values (e.g., CustomerID). Data types of key columns in both tables must match exactly before merging.

— JOIN TYPES

Understanding Power Query Join Types

Selecting the appropriate relational join behavior for business requirements:



01

Left Outer (Default)

Keeps ALL rows from the first table and brings in matching rows from the second table. Standard for lookup enrichment.



02

Inner Join

Keeps ONLY rows where keys match in BOTH tables. Drops unmatched transactions or orphan customers.



03

Anti Join (Left Anti / Right Anti)

Returns ONLY rows in one table that have NO match in the other table. Perfect for auditing missing lookup IDs!

— DATA LIFECYCLE

Managing Query Refresh Behavior

Keeping analytical outputs updated as new source files arrive:



01

Refresh All

Re-runs every query step across the entire workbook. Updates all tables, Pivots, and connected charts in 1 click.



02

Refresh (Single Query)

Right-click a specific query in the queries pane -> Refresh. Re-executes ETL steps for only that selected table.



03

Refresh on File Open

Query Properties setting that automatically triggers ETL refresh whenever the Excel workbook is opened by users.

— ARCHITECTURE TREE

Query Dependencies & Lineage

Tracing how data flows from raw source queries to intermediate staging and final reports:



01

The Query Dependencies View

Navigate to View tab > Query Dependencies. Displays a graphical flowchart mapping every raw source file, staging query, merged result, and output table.

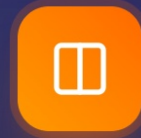


02

Why Lineage Inspection Matters

Helps analysts understand downstream impacts before deleting or editing a shared staging query. Prevents breaking dependent report tables.

Reshaping Data Layouts, Wide to Long Formats, Group By & Key Architectures



PART 2

Pivot & Unpivot Transformations

Reshaping Data Layouts, Wide to Long Formats, Group By & Key Architectures

Pivot vs Unpivot Concepts

The two fundamental data restructuring operations in modern data preparation:

 Interactive Unpivot Converter

Wide Layout (Un-analyzable)

Unpivoted Long Layout (Pivot Ready)

Raw Spreadsheet (Columns as Months)

4 Columns | 3 Rows

Product	Jan Sales	Feb Sales	Mar Sales
Laptop Pro	\$12,000	\$14,500	\$15,200
Wireless Mouse	\$3,200	\$3,800	\$4,100
USB-C Dock	\$5,400	\$6,100	\$5,900

KEY TAKEAWAY

Adding a new month column to a wide spreadsheet breaks all existing Pivot Table fields. Wide layouts cannot be filtered easily.

Power Query Command: Select static columns → Right Click → Unpivot Other Columns.

PRACTICAL UNPIVOT

Unpivoting Wide Spreadsheets

Why wide monthly column layouts break analytics and how Unpivot fixes them:

Interactive Unpivot Converter

Wide Layout (Un-analyzable)

Unpivoted Long Layout (Pivot Ready)

Raw Spreadsheet (Columns as Months)

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KEY TAKEAWAY

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— MATRIX GENERATION

The Pivot Column Operation

Converting normalized tall tables into custom cross-tabulation matrices:



01

When to Use Pivot Column

When you need to present summary totals side-by-side by category (e.g. pivoting metric names into separate columns for Revenue, Cost, Profit).



02

Configuring Aggregation Functions

Select the column to pivot -> click Pivot Column -> choose Value Column -> select Aggregation (Sum, Count, Average, Max, Min). Power Query computes matrix totals automatically.

— KEY IDENTIFICATION

Primary Key vs Foreign Key Architecture

Enforcing data integrity and relational matching across enterprise database tables:



01

Primary Key (Unique ID)

A column containing 100% unique identifiers with NO duplicates or nulls (e.g. EMP_ID in Employees table). Uniquely identifies every row.



02

Foreign Key (Reference ID)

A column in a child table (e.g. Projects table) that references the Primary Key in the parent table. Allows 1-to-Many relationships and guarantees data integrity.

— DATA SUMMARIZATION

Group By Summarizations

Collapsing thousands of detailed rows into concise aggregated category summaries:



01

Basic Grouping

Select a categorical column (e.g., Region) -> Group By -> compute Total Sales (SUM) or Transaction Count (COUNT).



02

Advanced Multi-Column Grouping

Group by multiple fields simultaneously (e.g. Region AND Product Category) to analyze multi-level breakdown metrics.

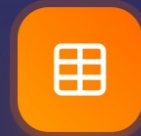


03

Multiple Aggregations

Add multiple summary columns in a single Group By step: Total Revenue (Sum), Average Order (Avg), Max Sale (Max), and Total Customers (Count Distinct).

Building Interactive Summaries, Value Field Settings, Custom Grouping & Calculated Fields



PART 3

Pivot Tables & Summarization

Building Interactive Summaries, Value Field Settings, Custom Grouping &
Calculated Fields

— FOUR QUADRANTS

The 4 Quadrants of a Pivot Table

Understanding the building blocks that control Pivot Table layout and calculations:



01

1. Filters Drop Zone

Top-level page filters that isolate specific sub-segments of the entire dataset (e.g. Year = 2024).



02

2. Columns Drop Zone

Categorical fields placed horizontally across the top to create matrix comparisons.



03

3. Rows Drop Zone

Primary categories displayed vertically down the left side of the summary table.



04

4. Values Drop Zone

Numeric metrics evaluated at row/column intersections (Sum of Sales, Count of Orders, Average Price).

— CREATION STEPS

Building Your First Pivot Table

A simple 3-step process to summarize thousands of raw data rows in seconds:



01

Step 1: Format as Excel Table

Select raw data -> press Ctrl + T. Formatting as an official Excel Table ensures your Pivot automatically expands when new rows are added.



02

Step 2: Insert Pivot Table

Navigate to Insert tab > PivotTable > choose New Worksheet. Excel creates a blank Pivot layout canvas and Field List pane.



03

Step 3: Drag & Drop Fields

Drag "Category" to Rows, "Region" to Columns, and "Sales Amount" to Values. Your summary table generates instantly!

— METRIC CONFIGURATION

Value Field Settings & Calculations

Customizing how numbers are aggregated and displayed inside your Pivot Table:



01

Summarize Values By

Switch between Sum, Count, Average, Max, Min, and Count Numbers depending on analytical objectives.



02

Show Values As

Display raw numbers as % of Grand Total, % of Column Total, % of Row Total, or Difference From Previous Month.



03

Number Formatting

Right-click metric -> Number Format -> apply Currency (\$) or Percentage (%) formatting to the underlying metric across the entire Pivot.

— HIERARCHICAL ANALYSIS

Grouping Data in Pivot Tables

Aggregating granular dates or numeric ranges into high-level analytical categories:



01

Date Grouping (Years, Quarters, Months)

Right-click any date cell in a Pivot Table -> click Group -> select Years, Quarters, and Months. Instantly transforms daily transaction logs into quarterly performance summaries!



02

Numeric Range Binning (Age / Income Bins)

Right-click numeric values (e.g. Customer Age) -> Group -> specify Start, End, and Interval (e.g. 20-29, 30-39, 40-49). Perfect for customer demographic distribution analysis.

— DERIVED METRICS

Calculated Fields in Pivot Tables

Adding custom mathematical formulas directly inside your Pivot Table without altering raw source data:



01

What is a Calculated Field?

A custom formula built inside the Pivot Table using existing fields (e.g., `Profit = Revenue - Cost` or `Commission = Sales \* 0.05`). Evaluates dynamically on aggregated totals.



02

How to Add a Calculated Field

Click inside Pivot Table -> PivotTable Analyze tab > Fields, Items, & Sets > Calculated Field -> type Name and Formula -> click Add. The custom metric appears in the Field List!

Chart Types, Visual Storytelling, Pivot Charts, Interactive Slicers & Timelines



PART 4

Charts & Dynamic Visualization

Chart Types, Visual Storytelling, Pivot Charts, Interactive Slicers & Timelines

— VISUAL ANCHORS

Visual Storytelling with Excel Charts

Transforming tabular rows into intuitive visual representations that highlight key patterns:



01

Clarify Trends & Patterns

Charts turn wall-of-number spreadsheets into instant visual insights that stakeholders comprehend in seconds.



02

Choose the Right Chart

Selecting the correct chart type (Column vs Line vs Scatter vs Histogram) ensures your message is accurate and unbiased.



03

Dynamic Interactivity

Combining Pivot Charts with Slicers and Timelines lets business users filter dashboards interactively during meetings.

— COMPARISON VISUALS

Column & Bar Charts

Representing categorical data with vertical or horizontal rectangular bars for direct comparison:



01

Column Chart (Vertical Bars)

Best for comparing discrete categories or short time series (e.g. Sales by Product Category or Quarterly Revenue). Height represents value.



02

Bar Chart (Horizontal Bars)

Ideal when category names are long (e.g. Country Names or Department Titles) or when ranking many categories from top to bottom.

— PART-TO-WHOLE

Stacked Column & Bar Charts

Dividing bars into sub-segments to display composition and parts of a whole within categories:



01

Stacked Column Chart

Bars are divided into colored segments representing sub-categories (e.g., Total Regional Sales split by Product Category: Electronics, Clothing, Decor).



02

100% Stacked Chart

Normalizes all bars to 100% height, emphasizing the relative percentage contribution of each sub-category regardless of total volume differences.

— PERCENTAGE BREAKDOWNS

Pie, Donut & Treemap Charts

Displaying percentage breakdowns and hierarchical category proportions:



01

Pie Chart

Displays slices of a circle representing proportions. Best when comparing 3 to 5 categories that sum to 100%.



02

Donut Chart

A modern variation of the pie chart with a hollow center. Leaves room for summary KPIs or total numbers inside the ring.



03

Treemap Chart

Displays hierarchical data as nested rectangles. Area size represents proportion, allowing visualization of many categories simultaneously.

— TREND ANALYSIS

Line & Area Charts

Connecting data points over time to emphasize continuous trends, seasonality, and direction:



01

Line Chart

Plots data points connected by lines over time (days, months, years). Essential for showing monthly sales trends, stock prices, or website traffic.



02

Area Chart

Similar to a line chart, but the space beneath the line is filled with color. Emphasizes total volume over time in addition to trend direction.

— CORRELATION & CLUSTERS

Scatter Plots & Correlation

Plotting individual data points to discover numeric relationships and spot outliers:



01

Scatter Plot (X vs Y)

Plots two continuous numeric variables on X and Y axes (e.g. Study Hours vs Exam Score or Age vs Salary). Reveals correlation strength and directional trends.



02

Types of Correlation

Positive Correlation (X up -> Y up), Negative Correlation (X up -> Y down), and No Correlation (random scatter). Trendline adds regression context.

— HISTOGRAM BINS

Histograms & Frequency Bins

Understanding numerical data distributions by grouping continuous values into range bins:



01

What is a Histogram?

A specialized column chart that shows the frequency distribution of continuous numerical data grouped into equal ranges (e.g. Age Bins: 20–29, 30–39, 40–49).



02

Why Histograms Matter

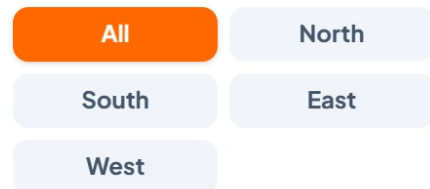
Helps analysts understand data spread, central tendency, skewness (left/right skew), bell-curve normality, and where the highest concentration of values lies.

INTERACTIVE VISUALS

Pivot Charts & Dynamic Dashboards

Creating visual charts linked directly to underlying Pivot Tables:

Region Slicer



Timeline Filter

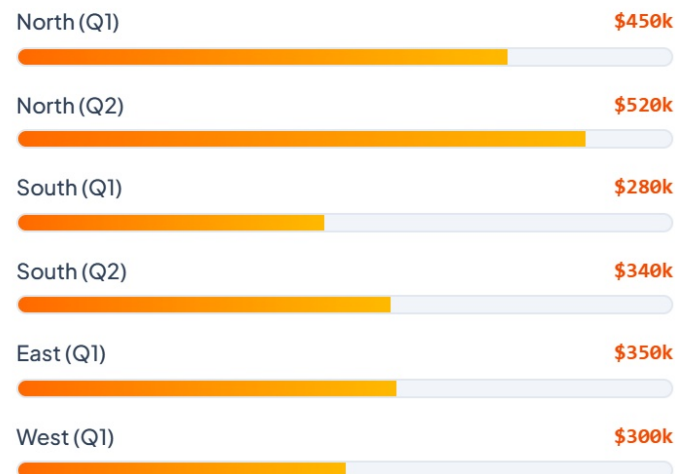


Live Pivot Output

Total: \$2240k

Region	Quarter	Sales (\$k)
North	Q1	\$450k
North	Q2	\$520k
South	Q1	\$280k
South	Q2	\$340k
East	Q1	\$350k
West	Q1	\$300k

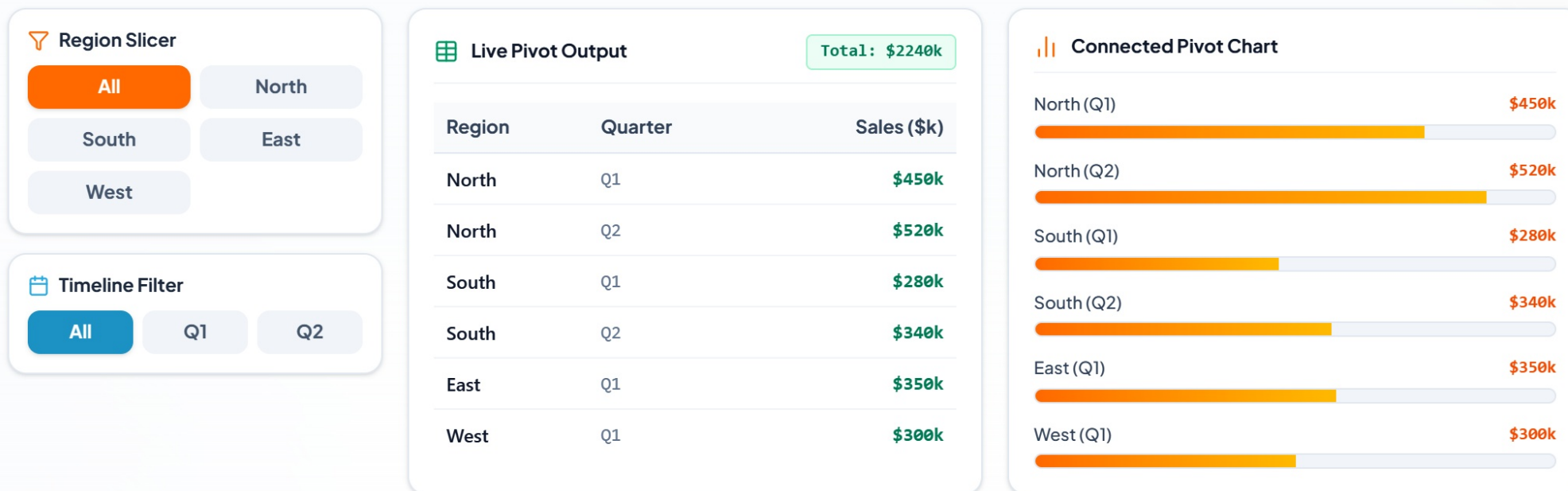
Connected Pivot Chart



— BUTTON FILTERING

Interactive Slicers

Visual button filters that allow users to filter tables and Pivot Tables with a single click:



— DATE NAVIGATION

Timeline Filters

Specialized visual time–slider controls designed exclusively for filtering date fields:

Region Slicer

All

North

South

East

West

Timeline Filter

All

Q1

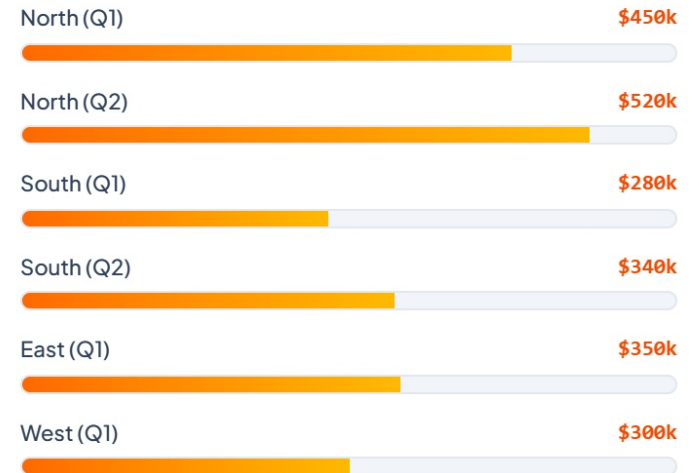
Q2

Live Pivot Output

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North	Q2	\$520k
South	Q1	\$280k
South	Q2	\$340k
East	Q1	\$350k
West	Q1	\$300k

Connected Pivot Chart



— MULTI-PIVOT LINKING

Connecting Slicers to Multiple Pivot Tables

The secret to building unified, fully interactive multi-chart executive dashboards:



01

Report Connections Window

Right-click Slicer -> Report Connections -> check the boxes for all Pivot Tables on your dashboard page.

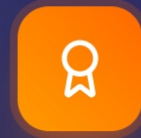


02

Unified Dashboard Interactivity

Clicking a single Slicer button (e.g., Region = "North") now filters ALL connected Pivot Tables, summary metrics, and Pivot Charts simultaneously across the dashboard!

5 Golden Rules, Common Pitfalls, Automated Regional Sales & Inventory Dashboard Case Study



PART 5

Best Practices & Real Business Project

5 Golden Rules, Common Pitfalls, Automated Regional Sales & Inventory Dashboard Case Study

— BEST PRACTICES

Golden Rules for Power Query & Pivots

Essential principles for building clean, maintainable, automated reporting models:

GOLDEN GUIDELINES

- ✓ Rename queries and transformation steps clearly (replace default "Changed Type1").
- ✓ Disable load for intermediate staging queries (right-click -> uncheck "Enable Load").
- ✓ Filter early, transform late (reduce row count first for maximum refresh speed).
- ✓ Document complex ETL steps with descriptive comments in Step Properties.
- ✓ Always format raw data as official Excel Tables (Ctrl + T) before building Pivots.

— ERROR PREVENTION

Common Pitfalls & How to Avoid Them

Avoiding common mistakes that break automated queries and reporting models:



01

Changing Source File Columns

Renaming or deleting source file column headers breaks downstream Power Query steps. Keep source headers standardized!



02

Mixing Text & Numeric Data Types

Keys stored as Text in Table A and Number in Table B cause Merge Queries to fail completely with zero matching rows.



03

Forgetting to Refresh Pivot Tables

Pivot Tables do not update automatically when raw source cells change. Always click Refresh All or enable Refresh on Open!

— END-TO-END PIPELINE

Capstone Project: Sales Analytics Pipeline

Combining Power Query ETL, Pivot Tables, and Pivot Charts into a live business dashboard:



01

1. ETL & Data Cleaning

Import monthly CSV sales files via Power Query, remove duplicate order IDs, unpivot regional month columns, and set exact Date types.



02

2. Relational Merge

Merge Sales Query with Product Lookup table on ProductID key (Left Outer Join) to pull Category and Unit Price columns.



03

3. Dashboard & Slicers

Build dynamic Pivot Tables for Monthly Revenue and Category Share. Add Slicers and Timelines connected to all Pivot Charts.

— SUMMARY

Key Takeaways from Session 06

Core principles to retain as you master advanced Excel, ETL, and data analytics:



01

Automates Repetitive Data Prep

Power Query handles messy CSVs, splits/merges, and cleanups in 1-click.



02

Unpivot Enables Analytics

Unpivoting wide spreadsheets into long format unlocks total Pivot Table flexibility.



03

Relational Key Matching

Merge queries connect transactions to lookup tables reliably via matching key IDs.



04

Instant Data Summarization

Pivot Tables aggregate millions of transactions in seconds with zero custom formulas.



05

Interactive Dashboard Slicers

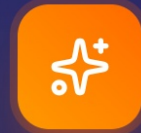
Slicers and Timelines turn static reports into dynamic visual dashboards.



06

End-to-End Data Mastery

Combining ETL + Pivots + Charts completes your core Excel analytics foundation!



From Raw Data to Insight.

Every Pivot Table, chart, and query transformation you learned today turns raw rows into decisions someone can act on.

Next session: Session 07 • SQL & Relational Databases (SELECT, WHERE, GROUP BY & JOINS)

Instant Academy • Data Analysis Training Program